

WHY THE MIXTURE FAILURE RATE BENDS DOWN WITH TIME

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Summary: It is well known that mixtures of decreasing failure rate (DFR) distributions are always DFR. It turns out that very often mixtures of increasing failure rate distributions (IFR) can decrease at least in some intervals of time. Usually this property can be observed asymptotically as $t \rightarrow \infty$. On the other hand, some rather stringent conditions can be formulated, which guarantee the preservation of the IFR property under the operation of mixing. In both cases the mixture failure rate is bent down compared with the corresponding unconditional expectation, which was proved previously for some specific cases. We generalize this result and discuss the “weakest populations are dying first” property, which leads to the change in the failure rate shape.

1. Introduction

Mixtures of decreasing failure rate (DFR) distributions are always DFR. It turns out that very often mixtures of increasing failure rate (IFR) distributions can decrease at least in some intervals of time. It means that the operation of mixing can change the pattern of aging. This fact is rather surprising and should be taken into account in various applications.

On the other hand, the mixture of IFR distributions can still preserve this property (Lynch, 1999). However, for both of these cases the mixture failure

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rate is bent down with time (we shall prove it in Section 3) in a sense, which will be specified after introducing some notation and discussing preliminary results in sections 2 and 3.

One can hardly find homogeneous populations in real life (reliability, survival analysis, demography) and mixtures of distributions usually present an effective tool for modelling heterogeneity. The natural approach for this modelling exploits a notion of a random unobserved parameter Z , which is usually called “frailty” (this term was introduced by Vaupel et al. (1979) for demographic applications). The foregoing leads to considering the random failure rate $\lambda(t, Z)$. As the failure rate is a conditional characteristic, the ‘ordinary’ expectation $E[\lambda(t, Z)]$ with respect to Z does not define a mixture failure rate $\lambda_m(t)$ and the proper conditioning should be performed.

A perfect experiment, illustrating the phenomenon mathematically described in this note, is performed by nature. It is well known that the mortality rates of humans obey the Gompertz lifetime distribution (Gompertz, 1825) with exponentially increasing failure rate (mortality rate). Assuming the proportional frailty model, which describes heterogeneity of human population:

$$\lambda(t, Z) = Z\alpha \exp\{\beta t\} \quad (1)$$

where α and β are positive constants, it can be shown (Finkelstein and Esaulova, 2001) that for some values of α and β (and for the gamma mixing distribution) the mixture failure rate $\lambda_m(t)$ is increasing in $[0, \infty)$ and asymptotically tends to a constant as $t \rightarrow \infty$. This fact explains recently observed deceleration of human mortality for oldest old (human mortality plateau, as in Thatcher, 1999) via the bending down effect to be properly defined in the next section.

The DFR case is trivial and we shall assume in what follows that $\lambda(t, z)$, $z \in [a, b]$ is increasing in t .

2. Definition of the bending down property

Let $T \geq 0$ be a lifetime random variable with the Cdf $F(t)$ ($\bar{F}(t) \equiv 1 - F(t)$). Assume that $F(t)$ is indexed by a random variable Z :

$$P(T \leq t | Z = z) \equiv P(T \leq t | z) = F(t, z)$$

and that the pdf $f(t, z)$ exists. Then the corresponding failure rate $\lambda(t, z)$ is $f(t, z)/\bar{F}(t, z)$. Let Z be interpreted as a non-negative random variable with support in $[a, b]$, $a \geq 0$, $b \leq \infty$ and the pdf $\pi(z)$. We can also describe $\lambda(t, Z)$ as a hazard rate process (Kebir, 1999). Thus, the mixture Cdf is defined by

$$F_m(t) = \int_a^b F(t, z) \pi(z) dz.$$

However, as the failure rate is a conditional characteristic, the mixture failure rate $\lambda_m(t)$ should be defined in the following way (see, e.g., Finkelstein and Esaulova, 2001):

$$\lambda_m(t) = \frac{\int_a^b f(t, z) \pi(z) dz}{\int_a^b \bar{F}(t, z) \pi(z) dz} = \int_a^b \lambda(t, z) \pi(z | t) dz, \quad (2)$$

where the conditional pdf (on condition that $T > t$):

$$\pi(z | t) \equiv \pi(z | T > t) = \pi(z) \frac{\bar{F}(t, z)}{\int_a^b \bar{F}(t, z) \pi(z) dz}. \quad (3)$$

On the other hand, consider the following **unconditional characteristic**

$$\lambda_p(t) = \int_a^b \lambda(t, z) \pi(z) dz, \quad (4)$$

which is, in fact, the unconditional expectation of the random hazard rate $\lambda(t, Z)$. It is clear that $\lambda_m(0) = \lambda_p(0)$. The function $\lambda_p(t)$ is a supplementary one, but it captures the monotonicity pattern of the family $\lambda(t, z)$ (under natural the condition of monotonicity in z for each t to be discussed in Section 4). Therefore, $\lambda_p(t)$ usually has a similar to individual $\lambda(t, z)$ shape: if, e.g., $\lambda(t, z)$, $z \in [a, b]$ is increasing, then $\lambda_p(t)$ is increasing as well. On the other hand, due to conditioning, $\lambda_m(t)$ can have a different pattern: it can ultimately decrease, for instance, or preserve the increasing in t property. However, it will be proved in Section 4, that

$$\lambda_m(t) < \lambda_p(t), \quad t > 0 \quad (5)$$

and under an additional assumption, that

$$(\lambda_p(t) - \lambda_m(t)) \uparrow, \quad t \geq 0. \quad (6)$$

Specifically, when

$$\lim_{t \rightarrow \infty} \lambda(t, z) = \infty, \quad z \in [a, b]$$

the difference in (6) also tends to ∞ .

Definition. *Relation (5) defines the weak bending down property for the mixture failure rate, whereas relation (6) defines the corresponding strong bending down property.*

3. Supplementary results

Consider the following specific multiplicative model

$$\lambda(t, z) = z\lambda(t), \quad (7)$$

where, $\lambda(t)$ is a baseline failure rate. This setting defines the frailty model which is widely used in applications. Equation (1), e.g., is a specific case of this model. Applying relation (2):

$$\lambda_m(t) = \int_0^{\infty} \lambda(t, z) \pi(z | t) dz = \lambda(t) E[Z | t] \quad (8)$$

and the conditional expectation $E[Z | t]$ plays a crucial role in defining the shape of the mixture failure rate $\lambda_m(t)$. The following result was proved in Finkelstein and Esaulova (2001):

$$E'_t [Z | t] = -\lambda(t) \text{Var}(Z | t) < 0,$$

which means that the conditional expectation of Z for the multiplicative model is a decreasing function of $t \in [0, \infty)$. On the other hand, (4) turns to

$$\lambda_p(t) = \int_a^b \lambda(t, z) \pi(z) dz = \lambda(t) E[Z | 0]. \quad (9)$$

As $E[Z | 0] \equiv E[Z]$ and $\lambda(t)$ is increasing, inequality (5) holds and

$$\lambda_p(t) - \lambda_m(t) = \lambda(t) (E[Z | 0] - E[Z | t])$$

is increasing. Therefore, in accordance with our definition the mixture failure rate is strongly bent down by operation of mixing. In the following illustrative example $\lambda(t)$ is increasing as a power function, but in spite of that, the mixture failure rate ultimately tends to 0.

Example 1. Let $\lambda(t, z) = 2zt$ and assume that $\pi(\theta)$ is a gamma probability density function ($a = 0$, $b = \infty$).

$$\pi(z) = \frac{\vartheta^\beta z^{\beta-1} \exp\{-\vartheta z\}}{\Gamma(\beta)}, \quad \beta, \vartheta > 0, \theta \geq 0.$$

The mixture failure rate can easily be obtained via direct integration:

$$\lambda_m(t) = \frac{2\beta t}{\vartheta + t^2}.$$

It is equal to zero at $t = 0$ and tends to zero as $t \rightarrow \infty$ with a single maximum at $t = \sqrt{\vartheta}$. Hence, the mixture of IFR distributions has a decreasing (tending to zero!) failure rate for sufficiently large t . Furthermore, the same result asymptotically holds for arbitrary Weibull distributions with increasing failure rate. ♦

This example shows that the mixture of the typical IFR distribution, obtained via the typical mixing distribution results in the non-monotone $\lambda_m(t)$, which even decreases to 0 as $t \rightarrow \infty$. However, some rather stringent conditions can be imposed to guarantee the IFR preservation property under operation of mixing. In Lynch (1999) the following result was obtained: *The mixture failure rate $\lambda_m(t)$ for increasing $\lambda(t, z)$ is increasing, if:*

1. $\bar{F}(t, z)$ is log concave in (t, z) ,
2. $\bar{F}(t, z)$ is increasing in z for each $t \geq 0$,
3. The mixing distribution is IFR.

The log-concavity is a natural, equivalent requirement for the IFR property of univariate distributions (Barlow and Proschan, 1975). Therefore, the first condition also seems natural for the bivariate case. The important stringent condition is, however, the second one. It is clear, that for the popular multiplicative (Cox proportional) model (7) this condition does not hold, as the survival function $\bar{F}(t, z)$ is decreasing in z for each $t \geq 0$ (larger values of z result in larger values of the failure rate). Therefore, the simplest candidate for the correspondent linear Weibull specific case, which meets the above conditions, would be

$$\lambda(t, z) = 2 \frac{t}{z}.$$

It can easily be seen that $\int_0^t \lambda(u, z) du$ for this specific case is a convex function of (t, z) (Block et al., 2003), but the practical application of this

inverse variation law is not so evident as that of the proportional law (7). The choice of the IFR mixing distribution is not so important. We shall come back to discussing these issues after formulating and proving the main result of this note.

4. The main result

Theorem. *Let the increasing in time failure rate $\lambda(t, z)$ in the continuous mixing model (2) be differentiable with respect to both arguments.*

Assume the following ordering in z :

$$\lambda(t, z_1) < \lambda(t, z_2), \quad z_1 < z_2, \quad \forall z_1, z_2 \in [a, b], \quad t \geq 0. \quad (10)$$

(a) *Then the mixture failure rate $\lambda_m(t)$ bends down with time at least in a weak sense.*

(b) *If additionally $\frac{\partial \lambda(t, z)}{\partial z}$ is increasing in t , then $\lambda_m(t)$ bends down with time in a strong sense.*

Proof. It is clear that ordering (10) is equivalent to the condition that $\lambda(t, z)$ is increasing in z for each $t \geq 0$. In accordance with equations (2) and (4) and integrating by parts (Finkelstein, 2004):

$$\begin{aligned} \Delta \lambda(t) &\equiv \int_a^b \lambda(t, z) [\pi(z) - \pi(z|t)] dz \\ &= \lambda(t, z) [\Pi(z) - \Pi(z|t)] \Big|_a^b - \int_a^b \lambda'_z(t, z) [\Pi(z) - \Pi(z|t)] dz \\ &= \int_a^b -\lambda'_z(t, z) [\Pi(z) - \Pi(z|t)] dz > 0, \quad t > 0, \end{aligned} \quad (11)$$

where

$$\Pi(z) = P(Z \leq z); \quad \Pi(z|t) = P(Z \leq z | T > t).$$

Inequality (11) and therefore the first part of the theorem follows from the assumption: $\lambda'_z(t) > 0$ and from the following inequality:

$$\Pi(z) - \Pi(z | t) < 0, \quad \forall t > 0, \quad z \in (a, b). \quad (12)$$

Inequality (12) can be interpreted via the principle: **“the weakest populations are dying out first”**, which is widely used, especially in the demographic literature (e.g., Vaupel, 2003). For obtaining (12) it is sufficient to prove that

$$\Pi(z | t) = \frac{\int_a^z \bar{F}(t, u) \pi(u) du}{\int_a^b \bar{F}(t, u) \pi(u) du}$$

is increasing in t , which will also be used for proving part b).

It is easy to see that $\Pi'_t(z | t) > 0$, if

$$\frac{\int_a^z \bar{F}'_t(t, u) \pi(u) du}{\int_a^z \bar{F}(t, u) \pi(u) du} > \frac{\int_a^b \bar{F}'_t(t, u) \pi(u) du}{\int_a^b \bar{F}(t, u) \pi(u) du}. \quad (13)$$

As $\bar{F}'_t(t, z) = -\lambda(t, z) \bar{F}(t, z)$, it is sufficient to prove that

$$B(t, z) \equiv \frac{\int_a^z \lambda(t, u) \bar{F}(t, u) \pi(u) du}{\int_a^z \bar{F}(t, u) \pi(u) du}$$

is increasing in z . Inequality $B'_z(t, z) > 0$ is equivalent to the following one:

$$\lambda(t, z) \int_a^z \bar{F}(t, u) \pi(u) du > \int_0^z \lambda(t, u) \bar{F}(t, u) \pi(u) du,$$

which is true, as $\lambda(t, z)$ is increasing in z .

Thus, due to additional assumption in b), the integrand in the end part of (11) is increasing and therefore $\Delta\lambda(t)$ as well, which immediately leads to the strong bending down property. ♦

Remark 1. We were considering rather crude sufficient conditions for proving the theorem. It seems that the condition that $\frac{\partial \lambda(t, z)}{\partial z}$ is increasing in t can probably be relaxed, but this needs further study.

Coming back to discussion of the IFR preservation property under operation of mixing, assume the inverse to (10) ordering

$$\lambda(t, z_1) > \lambda(t, z_2), \quad z_1 < z_2, \quad \forall z_1, z_2 \in [a, b], \quad t \geq 0 \quad (14)$$

and apply integration by parts to the right-hand side of equation (2):

$$\begin{aligned} \lambda_m(t) &= \int_a^b \lambda(t, z) \pi(z | t) dz \\ &= \lambda(t, b) - \int_a^b \lambda'_z(t, z) \Pi(z | t) dz. \end{aligned} \quad (15)$$

As $\lambda'_z(t) < 0$, there is a sum of two positive terms in the right-hand side of (15). In the same way as above it can be proved that now $\Pi(z | t)$ is decreasing in t . Therefore a crude sufficient condition for the preservation of the IFR property under operation of mixing can be formulated as the condition of increasing in t of the following product: $|\lambda'_z(t, z)| \Pi(z | t)$.

Example 2. Let $\lambda(t, z) = \lambda z^{-1}$, where λ is a constant. Therefore ordering (14) holds. Equation (15) for this specific case turns to

$$\lambda_m(t) = \frac{\lambda}{b} + \int_a^b \frac{\lambda}{z^2} \Pi(z | t) dz.$$

As $\Pi(z | t)$ is decreasing, $\lambda_m(t)$ is also decreasing confirming the general property that the mixture of decreasing (non-increasing) failure rates distributions is DFR irrespective of the mixing distribution and ordering in z . This illustrative example is important for the following remark.

Remark 2. It is clear that for the proportional model: $\lambda(t, z) = \lambda z$ the mixture failure rate is also decreasing. Assume now that $F(t, z)$ is a Weibull distribution with increasing failure rate. As follows from Example 1, the mixture failure rate $\lambda_m(t)$ for the proportional model (ordering (10)) increases for initial values of t and then tends to 0 as $t \rightarrow \infty$. On the other hand, for the same Weibull distribution but with different ordering in z , as was already mentioned in Section 3, $\lambda_m(t)$ is an increasing function in $[0, \infty)$. Thus the increasing in z ordering (10) bends down the mixture failure rate to a larger extent than decreasing ordering (14).

5. Concluding remarks

It turns out that the non-monotone pattern for the failure rate of mixtures of IFR distributions (usually ultimately decreasing for sufficiently large t) is more common than the corresponding preservation of the IFR property, for which rather stringent conditions should hold. In both cases, however, the mixture failure rate is bent down due to “the weakest populations are dying out first” effect, mathematically described in Section 4. This should be taken into account when analyzing the failure data for heterogeneous populations.

We’ve considered continuous mixtures. The shape of the resulting mixture failure rate for discrete distributions can be even more ‘irregular’ than for the continuous case. For instance, the mixture of two distributions with increasing linear failure rates can have four changes of monotonicity (Block et al., 2003).

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